

0 9

Read the following passage.

Carbohydrates are moved by mass transport in plants and in mammals. This movement is caused by a pressure gradient inside the transport vessels.

Plants transport most of their carbohydrate as sucrose. The phloem has sucrose-transport proteins in the cell-surface membrane. Potato plants with fewer sucrose-transport proteins produce fewer potatoes and accumulate sugars in their leaves.

5

Mammals do not have sucrose-transport proteins in their cell-surface membranes. They do have many monosaccharide-transport proteins in cell-surface membranes. These transport proteins allow monosaccharides through by facilitated diffusion.

10

Mammalian cells can change the number of glucose-transport proteins in their cell-surface membranes. The glucose-transport proteins are stored in internal membranes, then moved to the cell-surface membrane when they are needed. During exercise, there is a significant increase in glucose uptake by muscle cells.

15

Use the information in the passage and your own knowledge to answer the following questions.

0 9 . 1

Explain how sucrose-transport proteins in leaf cells enable the production of a pressure gradient in the phloem.

[3 marks]



0 9 . 2

Potato plants with fewer sucrose-transport proteins produce fewer potatoes and accumulate sugars in their leaves (lines 4, 5 and 6).

Explain why.

[2 marks]

0 9 . 3

During exercise, there is a significant increase in the glucose uptake by muscle cells (lines 14 and 15).

Give **two** reasons why glucose uptake by muscle cells increases significantly during exercise.

Explain your answers.

[4 marks]

1

2

Question 9 continues on the next page

Turn over ►



09.4

Which statement correctly describes the structure of both a sucrose-transport protein and a glucose-transport protein?

Tick (✓) **one** box.

[1 mark]

The secondary structure is held by disulfide bridges.
The tertiary structure allows the protein to be positioned on the inner surface of the cell membrane.

☐

The secondary structure is held by disulfide bridges.
The tertiary structure allows the protein to be positioned spanning the whole cell membrane.

☐

The secondary structure is held by hydrogen bonds.
The tertiary structure allows the protein to be positioned on the outer surface of the cell membrane.

☐

The secondary structure is held by hydrogen bonds.
The tertiary structure allows the protein to be positioned spanning the whole cell membrane.

☐

10

END OF QUESTIONS

